

Welcome to FSDS_GenAI_AgenticAL Programm-M Nagappa

1-Python Variable

```
In [7]: v = 9  
v
```

Out[7]: 9

```
In [2]: v1 = 63  
v1
```

Out[2]: 63

```
In [3]: nag = 39  
nag
```

Out[3]: 39

```
In [6]: v2 = 6  
v2
```

Out[6]: 6

```
In [8]: v2
```

Out[8]: 6

```
In [9]: v1  
v2  
nag
```

Out[9]: 39

```
In [10]: print(v1)  
print(v2)  
print(nag)
```

63

6

39

```
In [12]: import keyword  
keyword.kwlist
```

```
Out[12]: ['False',
          'None',
          'True',
          'and',
          'as',
          'assert',
          'async',
          'await',
          'break',
          'class',
          'continue',
          'def',
          'del',
          'elif',
          'else',
          'except',
          'finally',
          'for',
          'from',
          'global',
          'if',
          'import',
          'in',
          'is',
          'lambda',
          'nonlocal',
          'not',
          'or',
          'pass',
          'raise',
          'return',
          'try',
          'while',
          'with',
          'yield']
```

```
In [13]: len(keyword.kwlist)
```

```
Out[13]: 35
```

Variable_name = value

```
In [4]: age = 25 #Integer variable
        print(age)
```

25

```
In [8]: name = "nagu" # String Variable
        print(name)
```

nagu

```
In [9]: price = 39.99 #Float variable
        print(price)
```

39.99

```
In [10]: is_active = True #Boolean variable
         print(is_active)
```

True

Storing and Printing a Value

```
In [11]: x = 100    #printing the value of x
         print(x)
```

100

Using Variable in Expressions

```
In [13]: a = 3
         b = 6      #adding two variables and storing the result
         result = a + b
         print(result)
```

9

Changing the Value of a Variable

```
In [14]: score = 50  # initial value
         print(score)
```

50

```
In [15]: score = 100 # changing the value of score
         print(score)
```

100

Concatenating strings

```
In [18]: first_name = "M"      # Assigning values to variables
         last_name  = "Nagappa"

         full_name = first_name + " " + last_name    #concatenating string and storing in
         print(full_name)
```

M Nagappa

Using variable in a calculation

```
In [19]: length = 20    # assigning values to variables
         width  = 5

         area = length * width    # calculating the area of a rectangle
         print(area)
```

100

Reassigning Values to Variables

```
In [20]: x = 10    # initial value of x
```

```
print(x)
```

10

```
In [21]: x = 20      # Reassigning the value of x
print(x)
```

20

```
In [23]: # Variables in Python
```

```
first_name = 'M'
last_name = 'Nagappa'
country = 'India'
city = 'Bangalore'
age = 25
is_married = False
skills = ['HTML', 'CSS', 'JS', 'React', 'Python']
person_info = {
    'firstname': 'M',
    'lastname': 'Nagappa',
    'country': 'India',
    'city': 'Bangalore'
}
```

```
# Printing the values stored in the variables
```

```
print('First name:', first_name)
print('First name length:', len(first_name))
print('Last name: ', last_name)
print('Last name length: ', len(last_name))
print('Country: ', country)
print('City: ', city)
print('Age: ', age)
print('Married: ', is_married)
print('Skills: ', skills)
print('Person information: ', person_info)
```

```
# Declaring multiple variables in one line
```

```
first_name, last_name, country, age, is_married = 'M', 'Nagappa', 'India', 25, False

print(first_name, last_name, country, age, is_married)
print('First name:', first_name)
print('Last name: ', last_name)
print('Country: ', country)
print('Age: ', age)
print('Married: ', is_married)
```

```
First name: M
First name length: 1
Last name: Nagappa
Last name length: 7
Country: India
City: Bangalore
Age: 25
Married: False
Skills: ['HTML', 'CSS', 'JS', 'React', 'Python']
Person information: {'firstname': 'M', 'lastname': 'Nagappa', 'country': 'India', 'city': 'Bangalore'}
M Nagappa India 25 False
First name: M
Last name: Nagappa
Country: India
Age: 25
Married: False
```

In []: